

Week 4 – Workshop Plan

Gravity: Density, Superposition, and Geological Plausibility

Geophysics 3A: Potential Fields & Geothermics

March 10, 2026

Duration: 3 hours

Setting: Workshop room (whiteboards; projector)

Purpose

Strengthen intuition linking density contrasts and geometry to gravity anomalies, *without heavy derivations*. Students will use provided gravity solutions to reason about amplitude, width, and depth; apply *superposition* to build plausible anomalies from multiple bodies; and match geologic cross-sections to gravity profiles. The session primes students for the subsequent stairwell free-air practical and density measurements (Week 2 and lab).

Learning Outcomes

- Use given gravity solutions (slab, 2D cylinder, 2D polygon intuition) to reason about amplitude, width/half-width, and depth.
- Apply superposition to construct composite anomalies from multiple bodies with realistic $\Delta\rho$.
- Connect mineralogical density estimates to plausible $\Delta\rho$ for modeling and discuss detectability.
- Tackle non-uniqueness by proposing auxiliary observations that break model ties.

Reference: Week 4 notes on gravitational potential/fields, cylinder/polygon results, and width–depth relationships.

Roles and Grouping

Mixed groups of 4–5 with geology & physics/engineering. Rotate roles: *Facilitator, Math Lead, Geology Lead, Skeptic/QC, Recorder*.

Session Flow (180 min)

0–10 min: Warm-up — “Read the Curve”

Show three schematic gravity profiles: narrow positive peak; broad, low-amplitude swell; negative trough. In groups, identify sign, relative width (depth cue), and at least one plausible geologic cause for each. Share one observation per group.

10–35 min: Activity A — Density Reality Check

Goal: Ground $\Delta\rho$ used later in measured rock densities.

Given: Hand samples from Week 2 + their measured hydrostatic densities.

Tasks:

1. Estimate mineral fractions and compute theoretical density (weighted by mineral densities).
2. Compare to measured; compute % error; explain discrepancy (porosity, alteration, mis-ID).
3. Propose a realistic $\Delta\rho$ range (e.g., felsic vs. mafic, ore vs. host) to use in modeling today.
4. Comment on detectability of the resulting anomaly magnitude in a near-surface survey.

Deliverable: One-page table (*Density Worksheet*).

35–65 min: Activity B — Start-from-Solution Mini-Sprint

Goal: Use (not derive) standard results to reason about scale and shape.

Provided: *Formula Sheet* with: (i) infinite slab $\Delta g = 2\pi G \Delta\rho t$ and applicability; (ii) 2D cylinder *response shapes/width–depth cues*; (iii) polygonal body *edge/gradient* intuition.

Pick two micro-tasks:

1. Given target Δg and t , back-compute $\Delta\rho$ for a slab and sanity-check against Activity A.
2. From a pre-plotted cylinder response, measure half-width and estimate burial depth; predict how doubling $\Delta\rho$ or radius changes amplitude (not width).
3. For a simple 2D rectangle (polygon), mark where the steepest gradients occur and explain why edges dominate.

Deliverable: Board photo + “Assumptions & Limits” (3 lines).

65–110 min: Activity C — Superposition: Build-an-Anomaly

Goal: Construct a plausible local cross-section and its gravity by summing components.

Provided: *Superposition Component Library* with scaled kernels (normalized shapes) for: basement step (sedimentary wedge), mafic dike (2D cylinder proxy), felsic pluton (broad polygon), thin regolith cap (negative slab).

Tasks:

1. Draft a geologically plausible cross-section (2–3 bodies) with $\Delta\rho$ from Activity A.
2. Place laterally, set depths/widths, scale amplitudes, and *sum* the responses (trace overlays on graph paper).
3. Annotate why the geology is plausible (lithology, structure, relative timing).

Deliverable: Labeled cross-section + composite gravity + legend of component contributions.

110–150 min: Activity D — Match-Three: Cross-Sections ↔ Gravity Profiles

Goal: Practice non-uniqueness reasoning with realistic models.

Provided: *Match-Three Handout* — three local-style cross-sections (A–C) and three gravity profiles (1–3) designed to be confusable.

Tasks:

1. Match each profile to the most plausible cross-section using: (i) half-width/FWHM (depth cue); (ii) edge slopes/asymmetry (contacts, wedges); (iii) amplitude scaling with $\Delta\rho$ and size.
2. Propose one additional observation to break any tie (e.g., rock density control, mapping a contact, short-station-spacing gradient, borehole, susceptibility).
3. Role inversion: *Geologists* present the math-based width–depth rationale; *Phys/Eng* defend the geology.

Deliverable: One page with matches (A→?, B→?, C→?) + tie-breaker justification.

150–165 min: Practical Prep — Free-Air Gradient

The stairwell practical follows this workshop. In groups, draft a 3-bullet measurement plan (leveling, approach dial consistently to avoid backlash, clamp before moving, floor Δh , calibration factor, averaging and uncertainty). Target magnitude: $\sim 0.3086 \text{ mGal m}^{-1}$ (sign per course convention).

165–180 min: Exit Ticket

1. Which single assumption in your superposition model, if wrong, most alters your interpretation?
2. What extra observation would you collect next to reduce non-uniqueness, and why?

Materials

- Density Worksheet (Activity A).
- Formula Sheet (Activity B).
- Superposition Component Library (Activity C).
- Match-Three Handout (Activity D).
- Whiteboards, pens, graph paper, rulers.

Assessment

- Activity A worksheet (completeness; plausible $\Delta\rho$ ranges).
- Activity B board photo + assumptions.
- Activity C cross-section, composite gravity, and legend consistency.
- Activity D matches and quality of the tie-breaker argument.
- Exit ticket reflection.